PROJECT SUMMARY · AI-ASSISTED CLINICAL DECISION SUPPORT

IntelliCannula

NIR vein segmentation before peripheral IV cannulation, and daily photo-based monitoring of the insertion site afterwards.

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Problem

Peripheral IV cannulation often needs several attempts when veins are hard to see, and phlebitis at the insertion site is usually noticed only once it is visible to the eye. IntelliCannula assists the nurse at both stages. It advises and never acts: every output needs clinician confirmation.

System

A Next.js dashboard (also usable from a phone on the same Wi-Fi) talks to a FastAPI backend that runs both models, stores each patient's timeline and keeps an audit log of every prediction.

Before insertion: vein scan

Attention U-Net · CUBITAL NIR dataset · subject-level splits

Segments forearm veins in a near-infrared (850 nm) image and suggests an insertion site. Only verified NIR images are accepted, checked by where the image came from, so ordinary phone (RGB) photos are refused before the model runs.

After insertion: site check

YOLOv11 cannula detector · CIELAB redness · ORB scene matching

Checks photo quality, confirms a cannula is in frame, aligns the photo with the patient's day-1 baseline, and measures how much of the surrounding skin has become redder. Raises an alert when the change is large.

Measured results

COMPONENT	EVALUATION	RESULT
Vein segmentation	288 held-out test images	**Dice 0.567** (from 0.370) · IoU 0.413 · precision 0.495 · recall 0.758
NIR input check	RGB, grayscale and mislabelled images	0 reach the model
Cannula check	400 real images with no cannula	1–2% falsely accepted
Redness change	Real limb photos with known added redness (simulated)	**r 0.995** · mean error 3.6 pts · 0 / 27 false alarms
Speed	Warm requests, Apple GPU	vein scan 185 ms · site check 127–364 ms
Software quality	Automated tests	68 API · 192 module · 5 frontend · all pass

Limitations

- Vein model trained on paediatric NIR images (ages 5–22) from one camera rig.
- Cannula detector trained on 90 synthetic images; it misses some real cannulas.
- Multi-day site photos are simulated, so redness thresholds are not clinically validated.
- No NIR camera is attached; phone cameras cannot be used for vein scans.

Future work

- Connect a low-cost NIR imager (850 nm LEDs with an IR-pass camera).
- Ethics-approved collection of real IV-site photos with daily nurse VIP scores.
- Retrain the cannula detector on labelled real photos.
- Add login and encrypted storage before handling any real patient data.

Research prototype · not a medical device

Python · PyTorch · FastAPI · Next.js · OpenCV